

CariSurg MedTech Pathways Research Project

Preliminary Proposal

Project Title: Caribbean ED Validation of Machine-Learning Triage Support

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Statement of the Problem:

Emergency departments use structured triage systems to prioritise patients, but these systems still depend on staff judgement, local protocols, and available resources [7]. Recent AI triage studies report promising decision-support results for triage prediction, care-zone assignment, and patient prioritisation [1,3,4]. However, many models were developed in non-Caribbean settings using single-site, structured, or simulated data [3,4,6]. At Mercer General Hospital, this creates uncertainty about whether AI triage would be safe, fair, practical, and useful within Caribbean ED workflows and patient populations.

Previous Work:

Tyler et al. (2024) reviewed how artificial intelligence and machine learning are being used in emergency department triage. They found that machine learning models often performed better than conventional triage systems for predicting risk, hospital admission, disease identification, and resource needs. However, the review also showed that many studies relied on retrospective data, single-hospital datasets, and limited evidence on long-term patient outcomes [1].

Da'Costa et al. (2025) discussed how AI-driven triage systems may help emergency departments manage overcrowding, limited resources, and inconsistent patient prioritisation. Their review found that AI can support faster triage decisions, reduce waiting times, support staff, and improve resource allocation. However, they also highlighted challenges such as poor data quality, algorithmic bias, privacy concerns, accountability, limited clinician trust, and the need for more diverse datasets [2].

Araouchi and Adda (2024) developed TriageIntelli, an AI-assisted triage model using the Korean Triage and Acuity Scale. They compared several machine learning models, including SVM, Random Forest, ANN, GBM, Logistic Regression, XGBoost, and a stacking model. The stacking model performed best, with 80.05% accuracy, but the authors noted that the model relied on a single dataset, which limits its generalisability to other hospitals, patient groups, and clinical environments [3].

Kang et al. (2025) developed a predictive model to assign emergency department patients to care zones using data from Sengkang General Hospital in Singapore. Their model used triage data such as vital signs, demographics, NEWS scores, and nurses' triage notes. The Light Gradient-Boosting Machine model performed best, with an AUC of 0.95 on the holdout test set and 0.94 on a later clinical validation set. However, the model depended on local data and would need live testing, more diverse data, and continuous updating before wider use [4].

Alomari et al. (2025) tested whether ChatGPT could support emergency department triage using the Canadian Triage and Acuity Scale. ChatGPT showed high agreement with ED physicians at

85.61%, but its accuracy compared with a senior ED consultant was only 42.86%. The authors noted that the study was limited by its small sample size, the use of a general AI model not specifically trained for medical triage, and possible effects from case complexity, missing data, and physician experience [5].

Wang et al. (2025) studied whether large language models could support patient triage and guide patients to the correct department. After prompt engineering, GPT-4-Turbo achieved 100% MEWS-based triage accuracy, while GPT-4o achieved 96.20% and also reached 92.63% accuracy for outpatient department guidance. However, the study used Chinese patient scenarios, Chinese department structures, and structured descriptions, so the results may not fully apply to other healthcare systems [6].

Yancey and O'Rourke (2023) provided background information on modern emergency department triage methods. This background source explains that triage is used to sort and prioritise patients based on severity, urgency, and available resources. It also describes common triage systems such as START, Jump-START, the Emergency Severity Index, the Australasian Triage Scale, the Canadian Triage and Acuity Scale, and the Manchester Triage System. This source helps frame the proposal because the aim is not to replace existing triage methods, but to test whether machine learning can work beside them as a support layer [7].

Two main gaps were identified from the literature. The first gap is limited validation in Caribbean emergency department settings. Several papers showed strong AI triage performance, but many were tested in one country, one hospital, or one triage system [3,4,6]. This means Mercer cannot assume that these models would perform the same way for Caribbean patients, hospitals, resources, and workflows.

The second gap is limited evidence on how AI can fit into existing triage workflow. Many studies focus on model accuracy but less is said about whether doctors and nurses could use these tools easily during normal ED work [1,2,5]. This matters because an AI tool should support clinical judgement and improve decision-making, not create extra steps that slow staff down.

Proposed Solution:

To address the lack of Caribbean validation and the limited evidence on workflow integration, this project proposes a 12-week pilot to test a simple machine-learning triage support tool using de-identified Mercer ED data. The tool would not replace nurses, doctors, or current triage systems. Instead, it would work as a support layer beside existing triage methods, helping staff identify high-risk patients while keeping the process familiar and practical.

The pilot would use routinely collected triage information such as age, sex, vital signs, chief complaint, and existing triage category. A simple baseline model, such as logistic regression or a decision tree, would be created first. If time allows, this model could be compared with a stronger machine-learning model, such as Random Forest or Gradient Boosting.

The main aim would be to test whether the model can improve high-risk patient identification while keeping over-triage at an acceptable level. The pilot would measure high-acuity recall, over-triage rate, and agreement with existing triage outcomes. A simple workflow mock-up would also be created to show how the tool could fit beside current triage methods without disrupting doctors' and nurses' usual work.

This pilot would stay within a safe 12-week scope. It would not involve live hospital deployment, autonomous decision-making, regulatory approval, or identifiable patient data. Any future clinical use would require stronger validation, clinical oversight, and hospital or ethics approval.

References (5 to 10 *peer-reviewed journal or conference papers*):

References:

1. Tyler S, Olis M, Aust N, Patel L, Simon L, Triantafyllidis C, et al. Use of Artificial Intelligence in Triage in Hospital Emergency Departments: A Scoping Review. *Cureus*. 2024;16(5):e59906. doi:10.7759/cureus.59906.
2. Da'Costa A, Teke J, Origbo JE, Osonuga A, Egbon E, Olawade DB. AI-driven triage in emergency departments: A review of benefits, challenges, and future directions. *Int J Med Inform*. 2025;197:105838. doi:10.1016/j.ijmedinf.2025.105838.
3. Araouchi Z, Adda M. TriageIntelli: AI-Assisted Multimodal Triage System for Health Centers. *Procedia Comput Sci*. 2024;251:430–437. doi:10.1016/j.procs.2024.11.130.
4. Kang K, Annathurai A, Leow YK, Chia E, Fu Z. Development of Predictive Model for Patient Triage at Emergency Department. 2025 IEEE Conference on Artificial Intelligence (CAI). 2025:567–572. doi:10.1109/CAI64502.2025.00103.
5. Alomari LM, Alshammari MM, Arbaeen AO, Alshehri RA, Almalki HS. Safety and accuracy of AI in triaging patients in the emergency department. *Int J Emerg Med*. 2025;18:243. doi:10.1186/s12245-025-01069-x.
6. Wang C, Wang F, Li S, Ren QW, Tan X, Fu Y, et al. Patient Triage and Guidance in Emergency Departments Using Large Language Models: Multimetric Study. *J Med Internet Res*. 2025;27:e71613. doi:10.2196/71613.

Supporting Background Source:

7. Yancey CC, O'Rourke MC. Emergency Department Triage. In: StatPearls. Treasure Island (FL): StatPearls Publishing; 2023.